Information Extraction (intro + NER)

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Text Processing
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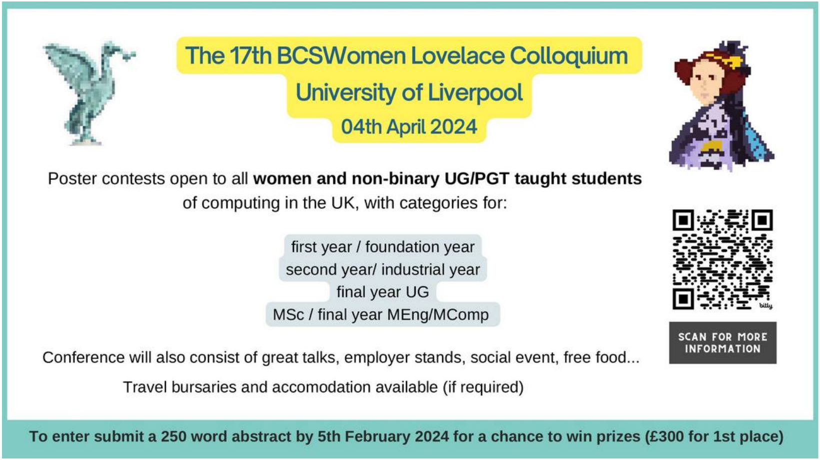
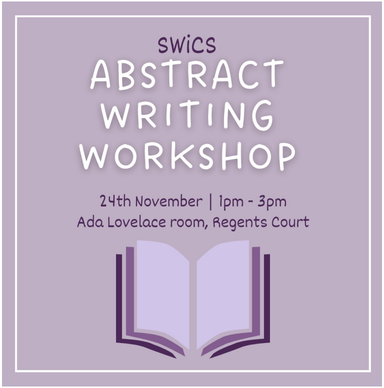

The 17th BCSWomen Lovelace Colloquium
University of Liverpool
04th April 2024
Poster contests open to all women and non-binary UG/PGT taught students of computing in the UK, with categories for:
first year / foundation year
second year/ industrial year
final year UG
MSc / final year MEng/MComp
Conference will also consist of great talks, employer stands, social event, free food...
Travel bursaries and accomodation available (if required)
SCAN FOR MORE INFORMATION
To enter submit a 250 word abstract by 5th February 2024 for a chance to win prizes (£300 for 1st place)


SWiCS
ABSTRACT WRITING WORKSHOP
24th November | 1pm - 3pm
Ada Lovelace room, Regents Court

Deadline: 01/12 @ 5pm

https://forms.gle/Ujm2kTjxLJxQGC9f7

CS Student Experience Video Competition

Do you have something interesting to say about your experience here in Sheffield?

To celebrate the diversity in community, we particularly invite students that are:

- Female / non-binary / LGBTQ+
- First of their family to attend University
- Part of the UoS Access+ programme

Submit a short video for a chance to win one of five £100 Amazon voucher!!!

Naive Bayes for Sentiment Analysis - Binary example

Consider the following **training set**:

doc	"good"	"poor"	"great"	(class)
d1.	3	0	3	pos
d2.	0	1	2	pos
d3.	1	3	0	neg
d4.	1	5	2	neg
d5.	0	2	0	neg

[Jurafsky and Martin, 2021]

Predict the sentiment of the text:

"A **good**, **good** plot and **great** characters, but **poor** acting."

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[Jurafsky and Martin, 2021]

Predict the sentiment of the text:

"A **good**, **good** plot and **great** characters, but **poor** acting."

Standard NB:

$p(\text{good}\mid\textbf{positive}) = 3/9 = 1/3$	$p(\text{good}\mid\textbf{negative}) = 2/14 = 1/7$
$p(\text{great}\mid\textbf{positive}) = 5/9$	$p(\text{great}\mid\textbf{negative}) = 2/14 = 1/7$

Naive Bayes for Sentiment Analysis - Binary example

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[Jurafsky and Martin, 2021]

Predict the sentiment of the text:

"A **good**, **good** plot and **great** characters, but **poor** acting."

Binary NB:

p(good|**positive**)	$= 1/4$	p(good|**negative**)	$= 2/6 = 1/3$
p(great|**positive**)	$= 2/4 = 1/2$	p(great|**negative**)	$= 1/6$

Exercise

Doc	Words	Class
1	A **sensitive**, **moving**, **brilliant** work	**Positive**
2	An **edgy** thriller that delivers a **surprising** punch	**Positive**
3	A **sensitive**, **insightful**, **flamboyant** film	**Positive**
4	Neither **revelatory** nor truly **edgy** – merely crassly **flamboyant** and comedically **labored**	**Negative**
5	**Unlikable**, **uninteresting**, **unfunny**, and completely, utterly **inept**	**Negative**
6	A sometimes **incisive** and **sensitive** portrait, targeting **sensitive** topics, that is undercut by its **awkward** structure	**Negative**
7	It's a sometimes **interesting** remake that doesn't compare to the **brilliant** original	**Negative**

Classify the following sentence using a **Binary Naive Bayes** classifier based on the above training data **without smoothing**.

Doc	Words	Class
8	A **flamboyant** romcom, **sensitive** but **awkward** at times.	???

Binary Naive Bayes for Sentiment Analysis - Exercise

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1. Calculate priors:

   p(**positive**) $= 3/7$ | p(**negative**) $= 4/7$

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1. Calculate priors:

   $p(\textbf{positive}) = 3/7 \mid p(\textbf{negative}) = 4/7$

2. Get likelihoods:

$$
\begin{array}{rl|rl}
p(\text{flamboyant}|\textbf{positive}) & = 1/8 & p(\text{flamboyant}|\textbf{negative}) & = 1/13 \\
p(\text{sensitive}|\textbf{positive}) & = 2/8 & p(\text{sensitive}|\textbf{negative}) & = \mathbf{1/13} \\
p(\text{awkward}|\textbf{positive}) & = 0/8 & p(\text{awkward}|\textbf{negative}) & = 1/13
\end{array}
$$

Binary Naive Bayes for Sentiment Analysis - Exercise

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8	A **flamboyant** romcom, **sensitive** but **awkward** at times.	?

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   p(flamboyant|**positive**) $= 1/8$ | p(flamboyant|**negative**) $= 1/13$
   p(sensitive|**positive**) $= 2/8$ | p(sensitive|**negative**) $=$ **1/13**
   p(awkward|**positive**) $= 0/8$ | p(awkward|**negative**) $= 1/13$

2. Compute posteriors

$$p(\textbf{positive}|text) = 3/7 * 1/8 * 2/8 * 0/8 = \textbf{0.0}$$

$$p(\textbf{negative}|text) = 4/7 * 1/13 * 1/13 * 1/23 = \textbf{0.00026}$$

Final decision: sentiment is **negative**

Learning objectives

- Name the main subtasks within Information Extraction
- Define the main approaches used for Information Extraction
- Explain the evaluation approaches for Information Extraction
- Explain NER
- Define the Knowledge Engineering Approach for NER

Information Extraction: Example

Who's News: @ Burns Fry Ltd.
04/13/94 WALL STREET JOURNAL (J), PAGE B10 ⟨CO⟩
BURNS FRY Ltd. (Toronto) – Donald Wright , 46 years old, was named executive vice president and director of fixed income at this brokerage firm. Mr. Wright resigned as president of Merrill Lynch Canada Inc. , a unit of Merrill Lynch & Co. , to succeed Mark Kassirer , 48, who left Burns Fry last month . A Merrill Lynch spokeswoman said it hasn't named a successor to Mr. Wright , who is expected to begin his new position by the end of the month .

- persons
- organisations
- locations
- times
- position in a company
- succession events

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Information Extraction: Filled template

<TEMPLATE-9404130062> :=
DOC_NR: "9404130062"
CONTENT: <SUCCESSION_EVENT-1>

<SUCCESSION_EVENT-1> :=
SUCCESSION_ORG: <ORGANISATION-1>
POST: 'executive vice president'
IN_AND_OUT: <IN_AND_OUT-1> <IN_AND_OUT-2>
VACANCY_REASON: OTH_UNK

<IN_AND_OUT-1> :=
IO_PERSON: <PERSON-1>
NEW_STATUS: OUT
ON_THE_JOB: NO

<PERSON-1> :=
PERSON_NAME: "Mark Kassirer"

<PERSON-2> :=
PERSON_NAME: "Donald Wright"

<ORGANIZATION-1> :=
ORG_NAME: "Burns Fry Ltd."
ORG_ALIAS: "Burns Fry"
ORG_DESCRIPTOR: "this brokerage firm"
ORG_TYPE: COMPANY
ORG_LOCALE: Toronto CITY
ORG_COUNTRY: Canada

<IN_AND_OUT-2> :=
IO_PERSON: <PERSON-2>
NEW_STATUS: IN
ON_THE_JOB: NO
OTHER_ORG: <ORGANIZATION-2>
REL_OTHER_ORG: OUTSIDE_ORG

<ORGANIZATION-2> :=
ORG_NAME: "Merrill Lynch Canada Inc."
ORG_ALIAS: "Merrill Lynch"
ORG_DESCRIPTOR: "a unit of Merrill Lynch & Co."
ORG_TYPE: COMPANY

Information Extraction vs. Information Retrieval

IR Task:

- Given a **document collection** and a **user query**
- Returns a (ranked) **list of documents relevant to the user query**

Information Extraction vs. Information Retrieval

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Strengths:

- Can search huge document collections very rapidly
- Insensitive to genre and domain of the texts
- Relatively straightforward to implement
  - challenges scaling to huge, dynamic document collections, e.g. the web

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Weaknesses

- Documents are returned rather than information/answers
  - user must further read texts to extract information
  - output is unstructured so limited possibilities for further processing
- Frequently not discriminative enough ("14,100,000 documents match your request")

Information Extraction vs. Information Retrieval

IE Task:

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- Returns a **structured representation** of all mentions of the specified entities, relations and/or events

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Strengths:

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- Can feed other powerful applications (databases, semantic indexing engines, data mining tools)

Weaknesses:

- Systems tend to be genre/domain specific and porting to new genres and domains can be time-consuming/requires expertise
- Limited accuracy
- Computationally demanding, so performance issues on very large collections

Information Extraction: example applications

Information Extraction: example applications

- Google News uses Named Entity Recognition for its "In the News" feature
- Scrapping web pages to build structured databases of job postings, apartment rentals, seminar announcements, etc.
- Assisting biomedical database curators by extracting biomedical entities and relations from the scientific literature prior to entry in a human-maintained database (e.g. Flybase)
- Assisting companies in competitor intelligence gathering, e.g. management or researcher succession events, new product or project annoucements, etc.

Information Extraction: Overview

- **Introduction to Information Extraction**
  - **Definition + contrast with IR**
  - **Example Applications**
  - **Overview of Tasks**
  - Overview of Approaches
  - Evaluation + Shared Task Challenges
  - Brief(est) history of IE
- Named Entity Recognition
  - Task
  - Approaches: Rule-based, Supervised Learning
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Information Extraction: Overview of tasks

Entity Extraction/Named Entity Recognition (NER)

Task: Identify the **extent** and the **type** of each textual mention of an entity

The **set of types** is determined in advance (e.g. organisation, person, date, etc...)

Information Extraction: Overview of tasks

Entity Extraction/Named Entity Recognition (NER)

Task: Identify the **extent** and the **type** of each textual mention of an entity

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Cable and Wireless today announced ...	Extent: 0-3 ; Type = ORG
IBM and Microsoft today announced ...	Extent: 0-1 ; Type = ORG
	Extent: 2-3 ; Type = ORG
John Lewis hired ...	Extent: 0-2 ; Type = ORG
Theresa May hired.	Extent: 0-2 ; Type = PER

Information Extraction: Entity extraction

Types of entities addressed by IE systems include:

- **Named individuals**
  - Organisations (ORG), persons (PER), books, films, ships, restaurants ...

  $\rightarrow$ Cable and Wireless today announced ...; Extent: **0-3** ; Type = **ORG**

  $\rightarrow$ Barack Obama was the 44th president ...; Extent: **0-2** ; Type = **PER**

Information Extraction: Entity extraction

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  - Geo-Political entities (GPE), locations (LOC)
  - → The Mont Blanc intersects France, Italy and Switzerland. ; Extent: **1-3** ; Type = **LOC**
  - → The Mont Blanc intersects France , Italy and Switzerland . ; Extent: **4-5** ; Type = **GPE**

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- **Times**: temporal expressions dates, times of day
  - → Let's meet at 2pm next Friday ...; Extent: **3-4** ; Type = **TIME**
  - → Let's meet at 2pm next Friday ...; Extent: **5-6** ; Type = **DATE**

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- **Measures**: monetary expressions, distances/sizes, weights ...
  - → This watch costs £35 ...; Extent: **3-4** ; Type = **MONEY**

Information Extraction: Entity extraction: coreference

Coreference

Different textual expressions that refer to the same real world entity are said to **corefer**. **Coreference Task**: link together all textual references to the same **real world entity**,

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Coreference

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Multiple references to the same entity in a text are **rarely made using the same string**:

- Pronouns: **Tony Blair** == **he**
- Names/definite descriptions: **Tony Blair** == **the Prime Minister**
- Abbreviated forms: **Theresa May** == **May**; **European Union** == **EU**
- Orthographic variants: **alpha helix** == **alpha-helix** == **α-helix** == **a-helix**

Can be seen as a separate task or as part of entity extraction task

Information Extraction: Relation extraction

Relation Extraction

Identify all assertions of relations between entities

Information Extraction: Relation extraction

Relation Extraction

Identify all assertions of relations between entities

May be divided into two subtasks:

- **Relation detection**: find pairs of entities between which a relation holds
- **Relation classification**: determine the type of a previously detected relation

Information Extraction: Relation extraction

Relation Extraction

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Example: LOCATION_OF holding between

  - ORGANISATION and GEOPOLOTICAL_LOCATION
  - medical INVESTIGATION and BODY_PART
  - GENE and CHROMOSOME_LOCATION

- EMPLOYEE_OF holding between PERSON and ORGANISATION
- PRODUCT_OF holding between ARTIFACT and ORGANISATION

Information Extraction: Relation extraction

Challenges for Relation Extraction

- Many different ways to **express the same relation**
  - **Canonical**: [Microsoft]$_{ORG}$ **is located in** [Redmond]$_{LOC}$
  - **Synonyms**: [Microsoft]$_{ORG}$ **is located/based/headquartered in** [Redmond]$_{LOC}$
  - **Syntactic variations**:
    - [Microsoft]$_{ORG}$, the software giant and ... , **is based in** [Redmond]$_{LOC}$
    - [Redmond]$_{LOC}$**-based** [Microsoft]$_{ORG}$ ...
    - [Redmond]$_{LOC}$**'s** [Microsoft]$_{ORG}$ ...
    - [Redmond]$_{LOC}$ software giant [Microsoft]$_{ORG}$ ...

Challenges for Relation Extraction

- Relations often involve **coreference links**

  [Bill Gates]$_{PER}$
  is co-founder, technology advisor and board member of [Microsoft]$_{ORG}$. **[He]**$_{PER}$ **served as chairman of the board** until Feb. 4, 2014. On June 27, 2008, **[Gates]**$_{PER}$ **transitioned out** of a day-to-day role in the company to spend more time on his global health and education work at the [Bill & Melinda Gates Foundation]$_{ORG}$.

Information Extraction: Event extraction

Event Extraction

Identify all reports of event instances, typically of a small set of classes

Information Extraction: Event extraction

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Identify all reports of event instances, typically of a small set of classes

May be divided into two subtasks:

- **Event detection**: find all mentions of events in a text
- **Event classification**: assign a class to the detected events

Examples

- National/european elections
- Management succession events
- Joint venture/product announcements
- Terrorist attacks

Information Extraction: Event extraction

Challenges for Event Extraction

Events may be simply viewed as relations. However they are typically complex relations

- often **temporally situated**, often short duration

→ Bolt etched his name in history with a 9.69-second finish **on Aug. 16, 2008**.

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→ 23 September 2019.
Banks, businesses, civil society and governments at all levels are to announce initiatives to finance and build a new generation of sustainable cities at the **UN Climate Action Summit** in New York today.

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- often expressed **across multiple sentences**

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Information Extraction: Overview of Approaches

Approaches to IE may be placed into four categories:

1. Knowledge Engineering Approaches
2. Supervised Learning Approaches
3. Bootstrapping Approaches
4. Distant Supervision Approaches

Information Extraction: Knowledge Engineering Approaches

[Mr. Wright]$_{PER}$, **[executive vice president]**$_{POSITION}$ of [Merrill Lynch Canada Inc.]$_{ORG}$

is-employed-by

Use **manually authored** rules that can be divided into

- "deep" – **linguistically inspired** language understanding systems
- "shallow" – systems engineered to the IE task, typically using **pattern-action rules**

Information Extraction: Knowledge Engineering Approaches

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Pattern: **"Mr. $Uppercase-initial-word"**
Action: **add-entity(person(Mr. $Uppercase-initial-word))**

Pattern: **"$Person, $Position of $Organization"**
Action: **add-relation(is-employed-by($Person,$Organization))**

Information Extraction: Supervised Learning Approaches

⇒ **Machine Learning systems** trained on manually annotated texts (entities and relations)

[Mr. Wright]$_{PER}$, **[executive vice president]**$_{POSITION}$ of [Merrill Lynch Canada Inc.]$_{ORG}$

is-employed-by

For each entity/relation create a training instance

- *k* words either side of an entity mention
- *k* words to the left of entity 1 and to the right of entity 2 plus the words in between

→ extract **features**: words, POS, morphology

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- patterns that match extraction targets
- classifiers that classify tokens as beginning/inside/outside a tag type

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Machine Learning techniques: covering algorithms, HMMs, SVMs

Information Extraction: Bootstrapping Approaches

$\Rightarrow$ A technique for relation extraction that requires only **minimal supervision**

Systems are given

- seed tuples: e.g. **⟨Microsoft , Redmond⟩**
- seed patterns: e.g. **[X]**$_{ORG}$ **is located in** **[Y]**$_{LOC}$

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→ process iterates until convergence

→ (detailed in a later lecture)

Information Extraction: Distant Supervision Approaches

⇒ also called **"weakly labelled"** or **"lightly supervised"** approaches

1. Assumes a (semi-)structured data source, such as
   - Wikipedia infoboxes (e.g. PERSON BORN__IN LOCATION/DATE)
   - Yeast Protein Database, (e.g. PROTEIN IS__LOCATED__IN SUBCELLULAR__ LOCATION)

   → contains tuples of entities standing in the relation of interest

   → eventually pointing to the source text

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3. **Labelled data is used to train a standard supervised NER or RE system**

   → See later lecture

Information Extraction: Overview

- **Introduction to Information Extraction**
  - **Definition + contrast with IR**
  - **Example Applications**
  - **Overview of Tasks**
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Information Extraction: Evaluation

How to evaluate an IE system?

Needs a fully annotated corpus (**test**) → produced manually

- Correct answers, called **keys**, or **reference** or **ground truth**
- ideally: several annotations by several annotators → **interannotator agreement**

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Principal metrics are:

- **Precision** (how much of what a system returns is correct)
- **Recall** (how much of what is correct a system returns)
- **F-measure** (a weighted combination of precision and recall)

Information Extraction: Evaluation Metrics

Confusion matrix for a **binary classification** task $\rightarrow$

Precision: how much of what system returns is correct

$$precision = P = \frac{TP}{TP + FP}$$

REFERENCE

HYPOTHESIS	+	-	Σ
+	**TP**	**FP**	HP
-	**FN**	**TN**	HN
Σ	RP	RN	**N**

Recall: how much of what is correct system returns

$$recall = R = \frac{TP}{TP + FN}$$

F-measure: a weighted combination of precision and recall

$$F_\beta = \frac{(1+\beta^2) * P * R}{\beta^2 * P + R} \Rightarrow F1 = \frac{2 * P * R}{P + R}$$

Information Extraction: Evaluation Metrics

Confusion matrix for a **multiclass classification** task $\rightarrow$ **Precision** and **Recall** for class i:

$$precision_i = P_i = \frac{R_i\_H_i}{H_i} \text{ and } recall_i = R_i = \frac{R_i\_H_i}{R_i}$$

Micro and **macro** averages

$$\text{Micro-}prec = \frac{\sum_i R_i\_H_i}{\sum_i H_i}$$

$$\text{Micro-}recall = \frac{\sum_i R_i\_H_i}{\sum_i R_i}$$

$$\text{Macro-}mes = \frac{\sum_i mes_i}{\#classes} \text{ with } mes \in prec, recall$$

REFERENCE

HYPOTHESIS	A	B	C	Σ
A	RA_HA	RB_HA	RC_HA	HA
B	RA_HB	RB_HB	RC_HB	HB
C	RA_HC	RB_HC	RC_HC	HC
Σ	RA	RB	RC	**N**

Shared Task challenges

Shared Task challenges or evaluation campaigns

competition among (international) research groups to build systems to address a specific task

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competition among (international) research groups to build systems to address a specific task

Conditions:

- clear **task definition**, annotated **text resources** for training, development and testing
- agreed **evaluation metrics** and **schedule**

→ results are usually discussed during a conference

Shared task challenges in IE include: Message Understanding Conferences (**MUC**, 1985-1998), Automatic Content Extraction (**ACE**, 1999-2008), Text Analysis Conference (**TAC**, 2008-2018), **BioCreative** (molecular biology, 2004-2016)

⇒ **Define the core methodology of the field and have led to significant progress**

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Brief(est) history of IE

1960s First published work on IE (though it was not called IE at the time)

1970s Precursor: the psychologist Roger Schank's work on scripts and story understanding

1980s Emergence of some commercial systems (financial transactions and newswires)
→MUC-1 in 1987

1990s MUC ran 7 times until 1998 and significantly advanced the field.
→Machine Learning approaches to IE began to appear

2000s ACE (1999-2008); succeeded by TAC (2008-present);

- → BioCreative (IE in the biomedical domain) began (2004-present);
- → work on IE in other languages began (e.g. Spanish, Japanese, Chinese, Arabic)

2010s TAC [`https://tac.nist.gov`] → **knowledge base population** track
⇒**Currently: a number of IE systems on the market & large, ongoing research effort**

[Information Extraction] Named Entity Recognition

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Information Extraction: NER - recap

Entity Extraction/Named Entity Recognition (NER)

Task: Identify the **extent** and the **type** of each textual mention of an entity

The set of types is determined in advance (e.g. organisation, person, date, etc...)

Cable and Wireless today announced ...	Extent: 0-3 ; Type = ORG
IBM and Microsoft today announced ...	Extent: 0-1 ; Type = ORG
	Extent: 2-3 ; Type = ORG
John Lewis hired ...	Extent: 0-2 ; Type = ORG
Theresa May hired.	Extent: 0-2 ; Type = PER

Information Extraction: NER - recap

Types of entities addressed by IE systems include:

- **Named individuals**
  - Organisations (ORG), persons (PER), books, films, ships, restaurants ...
  - → Cable and Wireless today announced ...; Extent: **0-3** ; Type = **ORG**
  - → Barack Obama was the 44th president... ...; Extent: **0-2** ; Type = **PER**
  - Geo-Political entities (GPE), locations (LOC)
  - → The Mont Blanc intersects France, Italy and Switzerland. ; Extent: **1-3** ; Type = **LOC**
  - → The Mont Blanc intersects France , Italy and Switzerland . ; Extent: **4-5** ; Type = **GPE**
- **Times**: temporal expressions dates, times of day
  - → Let's meet at 2pm next Friday ...; Extent: **3-4** ; Type = **TIME**
  - → Let's meet at 2pm next Friday ...; Extent: **5-6** ; Type = **DATE**
- **Measures**: monetary expressions, distances/sizes, weights ...
  - → This watch costs £35 ...; Extent: **3-4** ; Type = **MONEY**

Information Extraction: NER: coreference - recap

Coreference

Different textual expressions that refer to the same real world entity are said to **corefer**. **Coreference Task**: link together all textual references to the same **real world entity**,

Multiple references to the same entity in a text are rarely made using the same string:

- Pronouns: **Tony Blair** == **he**
- Names/definite descriptions: **Tony Blair** == **the Prime Minister**
- Abbreviated forms: **Theresa May** == **May**; **European Union** == **EU**
- Orthographic variants: **alpha helix** == **alpha-helix** == **α-helix** == **a-helix**

Can be seen as a separate task or as part of entity extraction task

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Information Extraction: Approaches to NER

Knowledge-engineering

- leverage linguistic resources created by experts

Supervised learning

- use of machine learning techniques

Bootstrapping

- Use of **seed patterns** to identify named entities
- Use known named entities to generate new patterns
- Rinse, repeat

Distant supervision / lightly supervised methods

- $\sim$ bootstrapping a machine learning system

Information Extraction: Approaches to NER

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NER: Knowledge Engineering Approaches

Dominant approach in the 1990s and still in use in many IE systems today.

Such systems typically use

- named entity **lexicons** and
- **manually authored** pattern/action rules or regular expression/FST recognisers

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Example: NER system, developed for participation in MUC-6

- described in Wakao et al. (1996) [Wakao et al., 1996]

→ recognizes **organisation**, **person**, **location** and **time** expressions in **newswire texts**

Another system: **GATE Annie** [Cunningham et al., 2014]
→Sheffield technology!
https://cloud.gate.ac.uk/shopfront/displayItem/annie-named-entity-recognizer

NER: Knowledge Engineering Approaches

System has four main stages:

1. Lexical processing
2. NE parsing
3. Discourse interpretation - Coreference Resolution
4. Discourse interpretation - Semantic Type Resolution

NER: Knowledge Engineering Approaches

1. **Lexical processing**

Rule-based NER systems use **specialized lexicons** $\rightarrow$ **gazetteers** (= geographical directory)

The Wakao et al. system has specialised lexicons for:

- **Organisations** – **2600** entries
- **Locations** – **2200** entries
- **Person names** – **500** entries
- **Company designators**: e.g. Corp, Ltd – **94** entries
- **Person titles**: e.g. Mr, Dr, Reverend – **160** entries

NER: Knowledge Engineering Approaches

1. **Lexical processing**

Rule-based NER systems use **specialized lexicons** → **gazetteers** (= geographical directory)

Why not use even larger gazetteers?

- Gazetteer of British Place Names containing over 50,000 entries
- → Many NEs occur in multiple categories
- → **The larger the lexicons the greater the ambiguity**
- Ex.: Ford ⇒ **company** or **person** or **place**
- listing of names is never complete → need a mechanism to type unseen NEs!

NER: Knowledge Engineering Approaches

1. **Lexical processing**

Example sentence: ”Norwich Investment Bank plc. today announced ...”

NER: Knowledge Engineering Approaches

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1a. **Tokenisation, sentence splitting, morphological analysis, Part-Of-Speech tagging**

$\rightarrow$ [Norwich]$_{NNP}$ [Investment]$_{NNP}$ [Bank]$_{NNP}$ [plc.]$_{NN}$ [today]$_{RB}$ [announced]$_{VBD}$ ...

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1b. **Gazetteer Lookup and Tagging:**

- **ORG**ganisations, **LOC**ations, **PER**sons, company designators (**CDG**), person titles

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1c. **Trigger Word Tagging**

- **trigger words** allow to classify certain multi-word names
- → Ex.: **Airlines** in "**Wing and Prayer Airlines**"
- system has trigger words for **ORG**anisations, **GOV**ernment institutions, **LOC**ations

→ [Norwich]$_{NNP/\mathbf{LOC}}$ [Investment]$_{NNP}$ [Bank]$_{NNP/\mathbf{ORG-TRIGGER}}$ [plc.]$_{NN/\mathbf{CDG}}$ [today]$_{RB}$ [announced]$_{VBD}$ ...

NER: Knowledge Engineering Approaches

2. **NE parsing**

Hand-produced rules:

- 177 for proper names
- 94 for organisation
- 54 for person
- 11 for location
- 18 for time expressions.

NER: Knowledge Engineering Approaches

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A fragment of the proper name grammar:

- NP –> ORGAN_NP
- NAMES_NP –> NNP NAMES_NP
- NAMES_NP –> NNP
- ORGAN_NP –> LIST_LOC_NP NAMES_NP CDG_NP
- ORGAN_NP –> LIST_ORGAN_NP NAMES_NP CDG_NP
- ORGAN_NP –> NAMES_NP '&' NAMES_NP

NER: Knowledge Engineering Approaches

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Rule "ORGAN_NP –> NAMES_NP '&' NAMES_NP" means:
an unclassified **proper name** (NAMES_NP) followed by '&' followed by an unclassified **proper name** is an **organisation name**
→ **Marks & Spencer** or **American Telephone & Telegraph**

NER: Knowledge Engineering Approaches

3. Discourse interpretation - **Coreference Resolution**

3a. When the name class of an **antecedent** (resp. **postcedent**) is known then establishing coreference allows the name class of the **anaphor** (resp. **cataphor**) to be established.

NER: Knowledge Engineering Approaches

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Anaphora/cataphora

In a narrower sense, **anaphora** is the use of an expression that depends specifically upon an **antecedent** expression and thus is contrasted with **cataphora**, which is the use of an expression that depends upon a **postcedent** expression. [Source: Wikipedia]

NER: Knowledge Engineering Approaches

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- Ex1.: **Ford Motor Co.** was founded in Detroit in 1903. **It** was the first to introduce...

NER: Knowledge Engineering Approaches

3. Discourse interpretation - **Coreference Resolution**

3b. An unclassified PN may be co-referential with a variant form of a classified PN, e.g.:

- Ex2.:
  - **Ford Motor Co.** was founded in Detroit in 1903, ..., **Ford** was the first to introduce...
  - **Creative Artists Agency** is a US talent agency ... In 2016, **CAA** had 1,800 employees

→ The unclassified PN may be inferred to have the same class as the classified PN.

→ Wakao et al. use 45 heuristics of this type for organisation, location, and person names.

NER: Knowledge Engineering Approaches

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3c. An unclassified PN may be co-referential with a definite NP

- Ex3.: **Kellogg**, the breakfast cereal **manufacturer**

NER: Knowledge Engineering Approaches

4. Discourse interpretation - **Semantic Type Inference**

Semantic type information about the arguments in certain **syntactic relations** is used to make inferences permitting the classification of PNs

4a. **noun-nous qualification**: PN qualifies an organisation-related object $\rightarrow$ organisation
   - $\rightarrow$ Erickson **stocks** $\Rightarrow$ **[Erickson]**$_{ORG}$

NER: Knowledge Engineering Approaches

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NER: Knowledge Engineering Approaches

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 → Miodrag Jones, **president of XYZ** ⇒ **[Miodrag Jones]**$_{PER}$

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4d. **verbal arguments**: PN names an entity involved in a verbal frame where the semantic type of the argument position is known → classify accordingly
   → Smith **retired from his position** as ⇒ **[Smith]**$_{PER}$

NER: Knowledge Engineering Approaches

Evaluation of Wakao et al.
MUC-6 NE evaluation set: a **blind test set** of 30 Wall Street Journal articles containing:

- 449 organisation names
- 373 person names
- 110 location names
- 111 time expressions

Results:

Proper Name Class	Recall	Precision	F1
Organisation	91 %	91 %	91.0 %
Person	90 %	95 %	92.4 %
Location	88 %	89 %	88.5 %
Time	94 %	97 %	95.5 %
Overall	91 %	93 %	92.0 %

⇒ Best system results on this evaluation had F1 measure = 96.42%
⇒ Human results were 96.68%

NER: Knowledge Engineering Approaches

Strengths

- **High performance** – only several points behind human annotators
- **Transparent** – easy to understand what system is doing/why

Weaknesses

- Porting to another domain requires substantial **rule re-engineering**
- Acquisition of **domain-specific lexicons**
- Rule writing requires high **levels of expertise**

Reading material

- Daniel Jurafsky & James H. Martin (2021).
  Chapter 17: Information Extraction.
  Speech and Language Processing, 3rd ed. draft.
  `https://web.stanford.edu/~jurafsky/slp3/17.pdf`

Credit: slides based on Dr Loïc Barrault's slides for the autumn 2020/2021 course.

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